ARMSTRONG

WDS-20

Dual Column Touchscreen Tensile Testing Machine



Product illustration. Confirm the supplied configuration and accessories in the quotation.

MODEL	CAPACITY / CONFIGURATION
WDS-20	500 kgf / approximately 4.90 kN

Servo-driven dual-column testing frame with TS7003L touchscreen, load cell, force and displacement display, and a micro-printer for test results.

Supplied and supported by Hung Ta Instrument.

Applications and specifications

For mechanical testing of rubber, plastics, plastic film, wire, cable, textiles, waterproofing and nonwoven materials, and suitable metal wire, foil, sheet and bar specimens. Use the correct grips and fixtures for tensile, compression, shear, tear, adhesion and 90/180-degree peel testing.

Parameter	Specification
Rated test load	500 kgf (approximately 4.90 kN)
Frame and drive	Dual-column frame; servo motor
Control	TS7003L touchscreen measurement and control system
Load resolution	1/25,000
Load accuracy	+/-1% as stated in the source; verify the configured range
Maximum travel	800 mm excluding fixtures
Test speed	1 to 500 mm/min; touchscreen input
Data sampling	10 Hz
Display	Current test value and peak value; built-in micro-printer
Selectable force units	kgf, N and lbf; available units depend on controller configuration
Machine dimensions	1010 x 490 x 1730 mm
Machine weight	Approx. 150 kg
Supply	Single phase, 220 V; technical table lists 6 A or as specified. Confirm the nameplate.

For quotation, provide the specimen material and size, test method, required force range, fixtures and report requirements. Warranty and service are governed by the Hung Ta quotation.

Technical reference

Operating and software guidance. Illustrations may show example readings or a different software revision. Use the instructions for the delivered unit.

2 Safety precautions

Observe all Danger, Warning and Caution instructions to prevent accidents and hazards when operating this instrument.

Signal word	Meaning
DANGER	Failure to comply may cause injury to the operator.
WARNING	Failure to comply may damage the instrument.
CAUTION	Failure to comply may affect test results and quality.
NOTE	Additional guidance for operating the product.

Safety symbols and machine appearance

Symbol	Meaning
Warning	Consult the operating manual where this symbol appears.
Hazardous voltage	High-voltage hazard.
Protective earth	Identifies the protective earthing terminal on the instrument.



Reference photograph. The actual configuration supplied takes precedence.

4 Operating instructions

Specimen preparation

Select specimens from the same batch of material. Cut and prepare them in accordance with the applicable standard.

Checks before testing

1. Confirm that the machine is connected to a power supply matching its specification.
2. Check that the limit-position switches and fixture screws are securely fastened.
3. Set the speed control to its minimum setting to prevent excessively fast movement and abnormal machine operation.
4. Securely clamp the test specimen in the grips.

Test procedure

1. Connect the power supply matching the machine nameplate.
2. Turn on the power switch.
3. Adjust the upper and lower limit collars to suitable positions and tighten them using a hex key.
4. Tap Up or Down to align the upper and lower grips.
5. Secure the specimen. Check that the display reads 0.00; otherwise, tap Zero. Select the required unit: kg, lb or N.
6. Tap Test to start testing. Refer to the controller instructions for the detailed test-function settings.
7. Repeat the procedure for the next specimen, or switch off the power.
8. Clean the machine.

Check the electrical rating on the machine nameplate before connection. The general operating example is not a wiring specification.

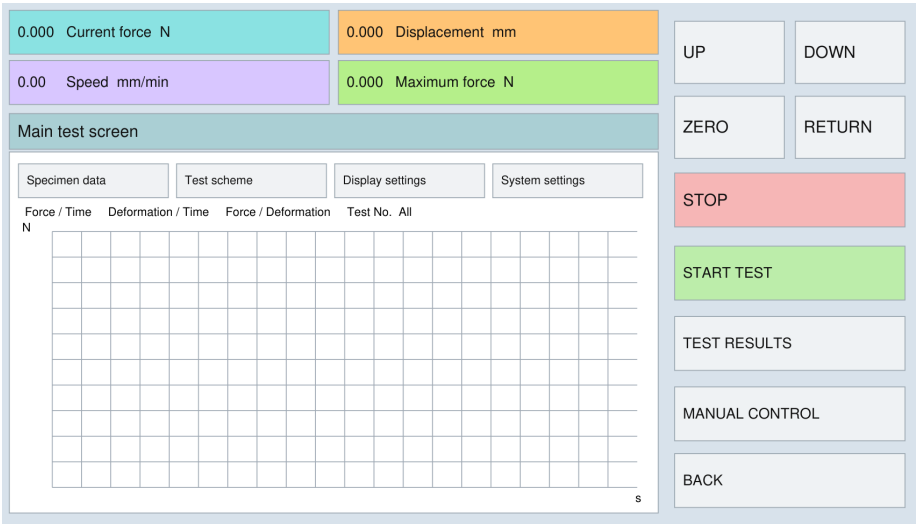
Touchscreen operation

Choose the required specimen and method, confirm travel limits, zero the readings and start the test. Contact Hung Ta for commissioning, wiring or calibration.

2.4 Starting a test and 2.5 Printing results

2.4 Starting a test

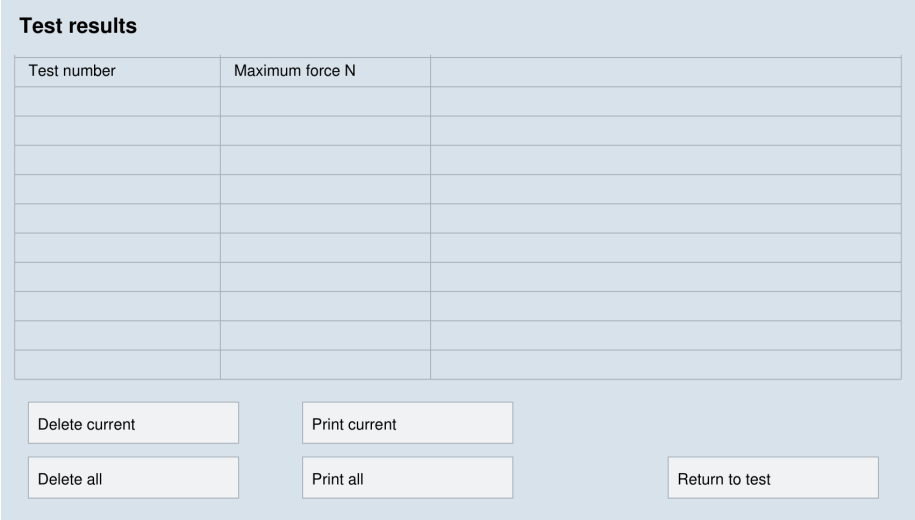
Check that the upper and lower limits will not interfere with the required test travel. Centre the specimen. Tap Start Test. The system automatically zeros the readings and begins the test. After the specimen collapses, the software returns the crosshead automatically and stops when the return time has elapsed. The test is then complete.



English reference for the main test screen

2.5 Printing results

Tap Test Results to view the current test count. Use the result-selection button in Test Scheme to select displayed results such as maximum force, deformation at maximum force, tensile strength, tear strength and peel strength. Print the current test result or all test results.



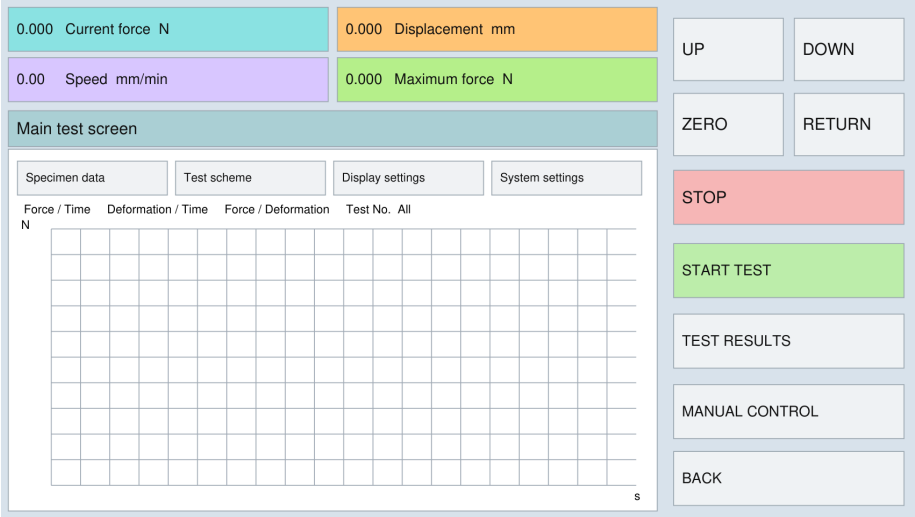
English reference for the test results screen

3.1 Entering the system

After normal power-up, tap Enter Test to open the system. Chinese and English are available. Select the language, then perform tensile or compression testing.

3.2 Main interface

The main interface contains a display area, a menu area and a control area.



English reference for the main interface

Main interface controls

Display area: Continuously updates the current force, maximum force, return time, test time and other measured values. A force-versus-time curve is displayed during a test.

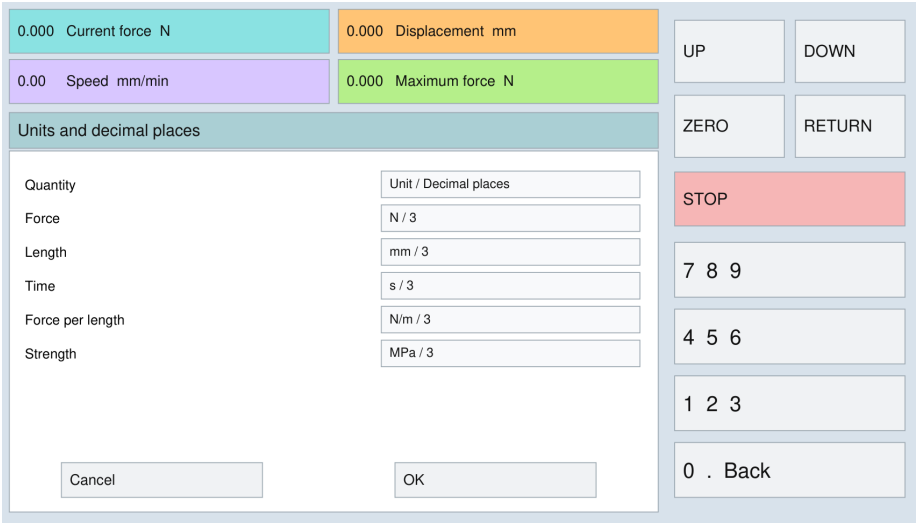
Menu area: Contains Specimen Data, Test Scheme, Display Settings and System Settings. Refer to the corresponding sections for details.

Control area: Contains the commonly used Start Test, Stop and Test Results buttons. The button functions are listed below.

Button	Function
Start Test	Starts a test. When the test finishes, testing ends and the crosshead returns automatically.
Stop	Ends the test or stops upward or downward movement.
Test Results	Opens the test results page.
Zero	Manually zeros force, time and other parameters.
Stop during a test	Stops the current action. If testing is active, the test stops and the crosshead returns automatically.
Up	Moves the crosshead upward.
Down	Moves the crosshead downward.
Return	Moves in the opposite direction to the last movement and stops when the return time has elapsed.
Manual Control	Opens the manual-control page to set the speed and perform jog movements.

3.3 Units and decimal places

Open Display Settings and select Units and Decimal Places.



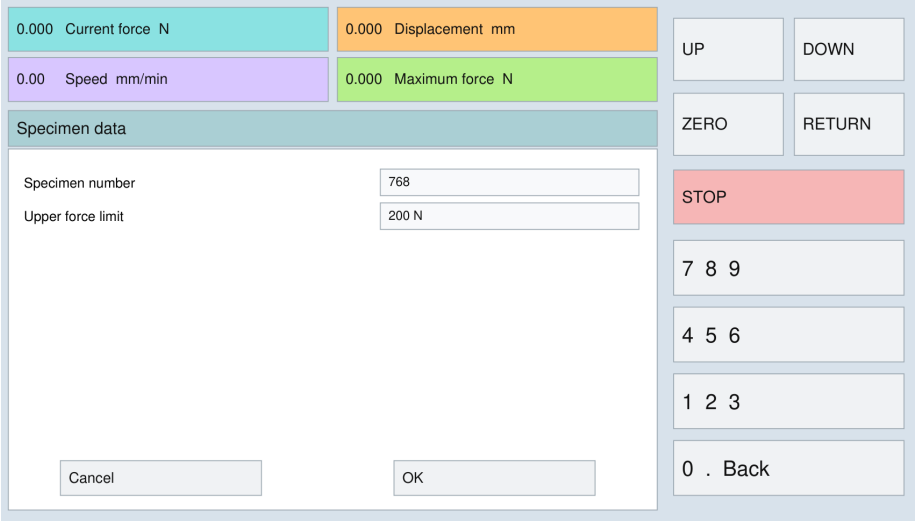
English reference for units and decimal places

Units and specimen data

Setting	Available selections
Force	gf, kgf, N, kN, tnf, lbf
Length	mm, inch
Time	s, min
Force per length	N/mm, N/cm, N/m, lbf/in, kgf/mm
Strength	Pa, KPa, MPa, kg/mm², kg/cm², kg/mm², kg/m², psi, psf
Decimal places	0, 1, 2 or 3

3.4 Specimen data settings

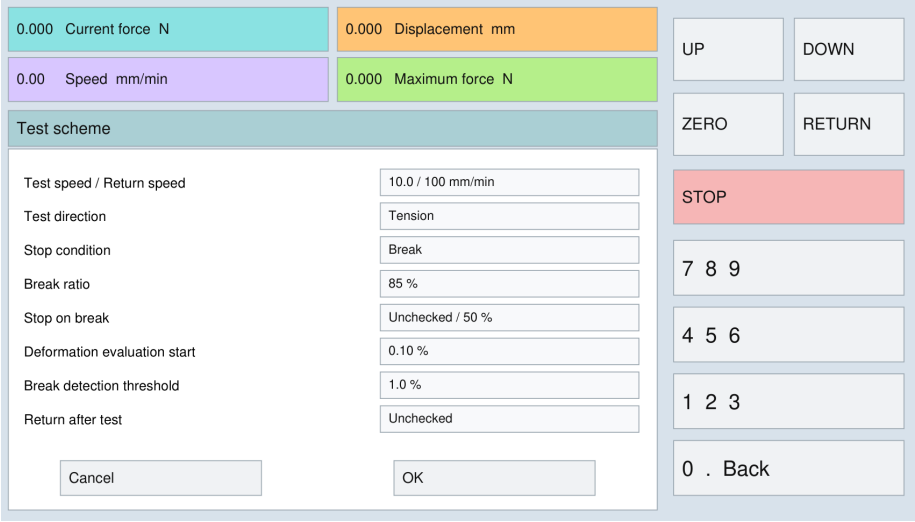
Tap Specimen Data to open the specimen-data settings page. Enter the specimen number and upper force limit.



English reference for specimen data

3.5 Test scheme

Tap Test Scheme to open the test-scheme settings page.

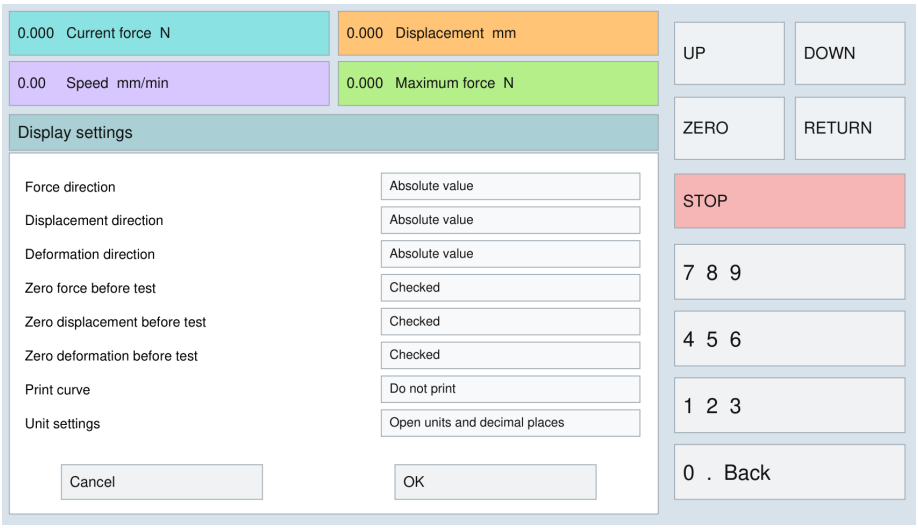


English reference for the test scheme

Setting	Function
Test speed	Operating speed after Start Test is selected.
Return speed	Operating speed during return.
Test direction	Select tension or compression.
Stop condition	Select Break, Deformation or Force.
Break ratio	Specimen-collapse break ratio. Default: 85%.
Stop on break	During a load-hold test, stops when specimen collapse is detected. Unchecked by default.
Deformation evaluation start	Starting point for deformation evaluation.
Break detection threshold	Initial force threshold for detecting specimen collapse. Default: 1% of load-cell capacity.
Return after test	Select whether the crosshead returns automatically after testing.
Force condition	Target force to be maintained during a load-hold test.
Hold time	Duration for which the force is maintained.
Deceleration factor	Slows movement during load holding to prevent excessive loading.
Load-hold start factor	Force-related factor for initiating load holding.

3.6 Display settings and 3.7 System settings

Tap Display Settings to open the display-settings page.



English reference for display settings

3.9 Test results

From the main interface, tap Test Results. View the results and print or delete the current result or all results. A maximum of 50 result sets can be recorded and displayed. When this limit is exceeded, the oldest result is removed automatically. Results are not retained after power-off; print them promptly after testing.



English reference for test results

5 Maintenance

1. Place the instrument on stable ground before use.
2. Do not move or handle the machine during operation.
3. Use the specified supply voltage. Excessive voltage may damage the components.
4. Contact our service department promptly if the instrument operates abnormally.
5. Ensure good ventilation while the machine is operating.
6. Lubricate mechanical components regularly because they operate under substantial loads.
7. Clean the machine after each test and keep it clean.

8. Wipe the control box with a dry cloth, not a wet cloth.

9. Ensure that the equipment is properly earthed.

Service and support

For technical support, provide the ARMSTRONG model, serial number and a description of the issue to Hung Ta Instrument.